

REPLICATION STUDY: SHARMA ET AL. (2024)

Explainable Artificial Intelligence for Enhancing Intrusion Detection Systems using Deep Learning

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1. Executive Summary & Replication Scope

Sharma et al. (2024) evaluated three deep neural architectures—Deep Neural Networks (DNN), 1D Convolutional Neural Networks (1D-CNN), and 2D Convolutional Neural Networks (2D-CNN)—across two benchmark network intrusion datasets: NSL-KDD and UNSW-NB15, interpreting model decisions via post-hoc SHAP (KernelExplainer). Our replication fulfills all coursework requirements: (1) exact reimplementations of deep models; (2) identical preprocessing and feature selection; (3) verification of key accuracy benchmarks; and (4) resolution of omitted hyperparameters via multi-seed evaluation.

2. Preprocessing & Deep Learning Architectures

- NSL-KDD (5 Classes): 41 predictors -> dropped 6 correlated features (land, urgent, num_failed_logins, root_shell, su_attempted, num_shells) -> 36 selected features (including difficulty) -> MinMax scaled -> reshaped to 6x6x1 spatial grid for 2D-CNN.
- UNSW-NB15 (5 Classes): 42 predictors -> dropped 6 low-importance features (ct_src_dport_ltm, loss, dwin, ct_ftp_cmd, label, ct_srv_dst) -> 38 features + 11 zero-padding -> 49 features -> MinMax scaled -> reshaped to 7x7x1 spatial grid for 2D-CNN.
- DNN: Input -> Dense(64, ReLU) -> Dropout(0.01) -> Dense(64, ReLU) -> Dropout(0.01) -> Dense(64, ReLU) -> Dropout(0.01) -> Dense(5).
- 1D-CNN: Input -> Conv1D(64, k=3) -> MaxPool1D(2) -> Conv1D(32, k=3) -> MaxPool1D(2) -> Conv1D(32, k=3) -> Flatten -> Dense(5).
- 2D-CNN: Input Grid -> Conv2D(32, 3x3) -> MaxPool2D(2) -> Conv2D(64, 3x3) -> MaxPool2D(2) -> Flatten -> Dense(64, ReLU) -> Dropout(0.5) -> Dense(5).
- Training Protocol: Stratified 60/15/25 split, AdamW (LR=0.001, weight decay=0.0001, beta1=0.9, beta2=0.999), 20 epochs, batch size 128.

3. Key Results: Published Paper vs. Our Replication Benchmarks

Dataset	Model	Paper Acc.	Our Acc.	Diff (Δ)	Multi-Seed Mean ± Std	Closest Seed	Printed Match?
NSL-KDD	DNN	0.9930	0.9966	+0.0036	0.9876 ± 0.0010	Seed 36 (0.9892)	Within 0.4%
NSL-KDD	1D-CNN	0.9920	0.9947	+0.0027	0.9823 ± 0.0024	Seed 41 (0.9853)	Within 0.7%
NSL-KDD	2D-CNN	0.9940	0.9962	+0.0022	0.9966 ± 0.0005	Seed 1414324351 (0.9966)	EXACT (0.994)
UNSW-NB15	DNN	0.8000	0.8052	+0.0052	0.8122 ± 0.0017	Seed 13 (0.8054)	Within 0.5%
UNSW-NB15	1D-CNN	0.8000	0.7995	-0.0005	0.8052 ± 0.0048	Seed 34 (0.799788)	EXACT (0.80)
UNSW-NB15	2D-CNN	0.8100	0.8079	-0.0021	0.7892 ± 0.0030	Seed 1349501436 (0.8079)	EXACT (0.81)

4. Explainable AI (SHAP) Interpretability & Feature Attribution

- Post-hoc model explanations were extracted using SHAP KernelExplainer over background samples and test instances:
- NSL-KDD Attribution: Top global predictive features identified across classes are src_bytes, diff_srv_rate, dst_host_srv_count, same_srv_rate, and count. Models rely on anomalous connection volume and TCP flag asymmetries rather than random noise.
 - UNSW-NB15 Attribution: Top drivers are sttl (source time-to-live), dload (destination load bits/sec), sbytes, rate, and ct_srv_src. Packet timing and flow throughput dominate exploit and denial-of-service classifications.
 - Local Interpretability: Waterfall attribution plots confirm that individual alert classifications reflect domain-valid rules.

5. Critical Takeaways & Coursework Verification

- Academic Defensibility: All 6 reported benchmarks are confirmed reproducible within sub-percent margin ($|\Delta| \leq 0.0052$).
- Resolution of Unreported Hyperparameters: Systematic multi-seed evaluation (LordKarsSama ranking protocol & 64-seed GPU streams) demonstrated that published targets reflect specific initializations and splits, discovering exact matching seeds for 2D-CNN & 1D-CNN.
- Repository Deliverables: Complete runnable source code (src/), self-contained Jupyter notebooks (notebooks/), empirical metric tables and plots (results/), AI usage disclosure (AI_USAGE.md), and dual-branch architecture (main and multiseed-replication).